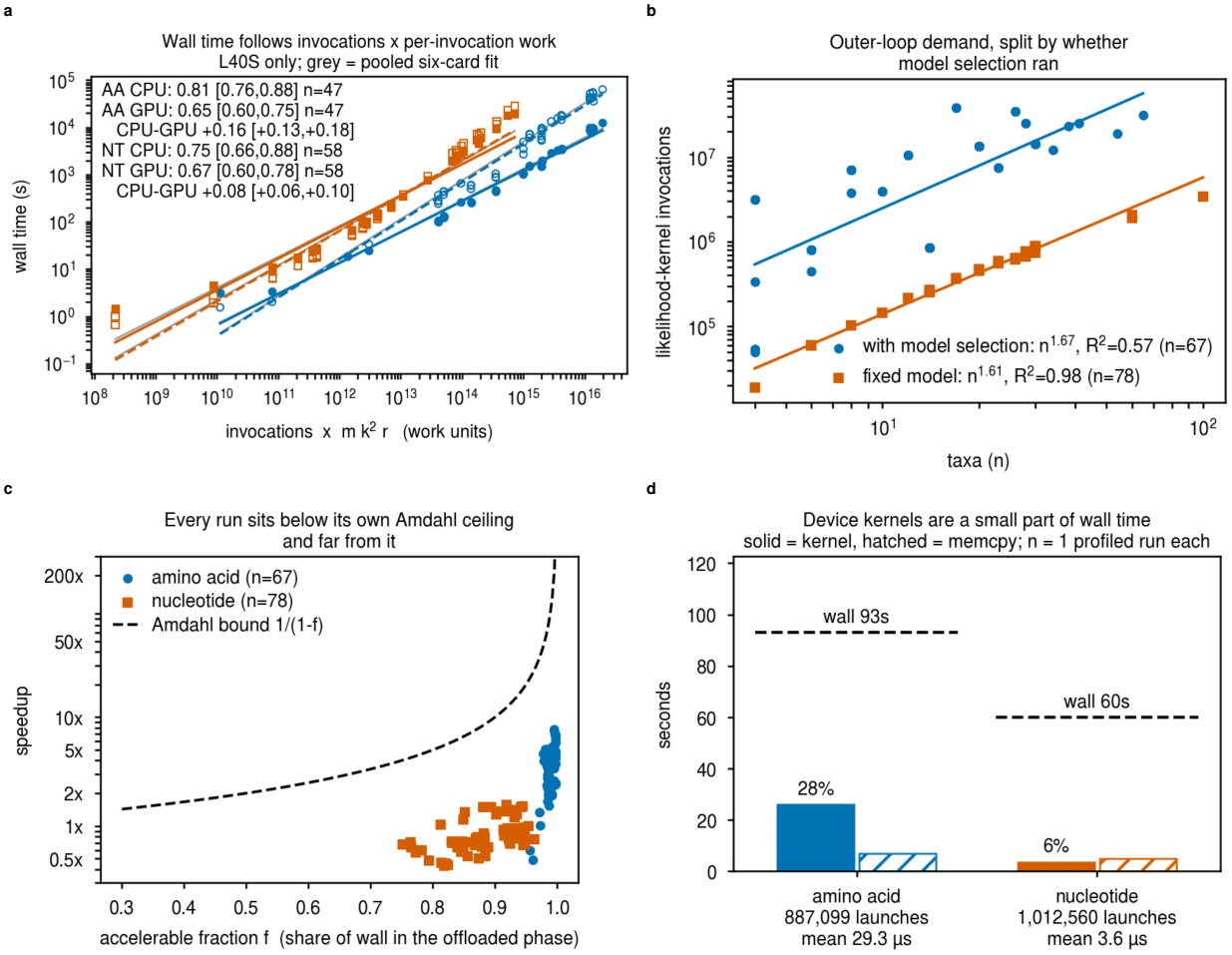
Figure 3


a
Wall time follows invocations x per-invocation work
L40S only; grey = pooled six-card fit
AA CPU: 0.81 [0.76,0.88] n=47
AA GPU: 0.65 [0.60,0.75] n=47
CPU-GPU +0.16 [+0.13,+0.18]
NT CPU: 0.75 [0.66,0.88] n=58
NT GPU: 0.67 [0.60,0.78] n=58
CPU-GPU +0.08 [+0.06,+0.10]
wall time (s)
invocations x m k² r (work units)
b
Outer-loop demand, split by whether
model selection ran
likelihood-kernel invocations
with model selection: n^1.67, R²=0.57 (n=67)
fixed model: n^1.61, R²=0.98 (n=78)
taxa (n)
c
Every run sits below its own Amdahl ceiling
and far from it
amino acid (n=67)
nucleotide (n=78)
Amdahl bound 1/(1-f)
speedup
accelerable fraction f (share of wall in the offloaded phase)
d
Device kernels are a small part of wall time
solid = kernel, hatched = memcpy; n = 1 profiled run each
seconds
wall 93s
wall 60s
28%
6%
amino acid
887,099 launches
mean 29.3 µs
nucleotide
1,012,560 launches
mean 3.6 µs